Fundamental of Software Project Management (Assignment 1)

Project Title:

National Freelance and Skill Verification Platform

Name:

Tanees Fatima

Roll Number:

23i-3003

Section:

BSE-6B

Department of Software Engineering

FAST – National University of Computer & Emerging Sciences

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2 Introduction / Background

Pakistan is one of the fastest-growing freelance economies in the world, with thousands of individuals offering services in software development, design, content writing, digital marketing, and other IT-enabled domains. Despite this growth, the freelance ecosystem faces major challenges related to trust, skill verification, payment security, and job matching. Many employers struggle to identify genuine talent, while skilled freelancers often fail to secure fair opportunities due to lack of verified credentials.

This project proposes the development of a National Freelance and Skill Verification Platform that provides a secure, transparent, and reliable digital environment for freelancers, employers, and institutions. The platform aims to formally verify skills, portfolios, and experience, enabling trusted collaboration and sustainable growth of the freelance economy in Pakistan.

3 Business Objectives

The primary objectives of this project are:

- To provide a trusted digital platform for verified freelancers and employers.
- To reduce fraud, fake profiles, and misrepresentation of skills.
- To improve employment opportunities for skilled Pakistani freelancers.
- To introduce standardized skill assessment and verification mechanisms.
- To support Pakistan's digital economy and global competitiveness.

This project aligns with the objectives of the Software Project Management course by enabling students to design and manage a large-scale, modular software system. It also aligns with the degree program by providing real-world exposure to project planning, system integration, and stakeholder coordination, while supporting professional growth through practical industry-oriented learning.

4 Current Situation and Problem / Opportunity Statement

4.1 Current Situation

Existing freelance platforms mainly rely on self-declared skills, unverified portfolios, and rating-based systems. Employers often face difficulties in evaluating true competence, while freelancers

suffer from delayed payments, unfair disputes, and limited credibility especially due to the rise of Artificial Intelligence.

4.2 Problems

- Fake or exaggerated skill claims
- Lack of standardized skill verification
- Trust issues between employers and freelancers
- Payment disputes and delayed transactions
- Limited visibility for genuinely skilled freelancers

4.3 Consequences of Inaction

If these issues remain unresolved, Pakistan risks losing high-value freelance opportunities, reduced global trust in its workforce, and continued exploitation of skilled professionals.

4.4 Opportunity

A centralized, verified, and secure platform can address these challenges while strengthening Pakistan's freelance ecosystem.

5 Critical Assumptions and Constraints

5.1 Assumptions

- Freelancers and employers are willing to adopt a verified platform.
- Skill assessments can be standardized and automated.
- Internet and smartphone access remain widely available.
- Government, universities, or private partners may support adoption.

5.2 Constraints

- Limited development time (one academic semester).
- Budget constraints typical of an academic project.
- Need to rely on existing open-source technologies.
- Legal and regulatory considerations related to payments and data privacy.

6 Analysis of Options and Recommendation

Option 1: Do Nothing

Continue relying on existing freelance platforms.

Disadvantage: Problems of trust, fraud, and unfair practices remain unresolved.

Option 2: Modify Existing Platforms

Enhance current platforms with minor verification features.

Disadvantage: Limited control, dependency on third-party systems.

Option 3: Develop a New National Platform (Recommended)

Design and develop a dedicated platform focused on skill verification, trust, and secure transactions.

Advantage: Full control, tailored features, long-term scalability.

Recommendation: Option 3 is the most suitable and sustainable solution.

7 Preliminary Project Requirements

7.1 Functional Requirements

- User registration and authentication (freelancers and employers)
- Skill assessment and certification module
- Portfolio and document verification
- Job posting and matching system
- Secure payment mechanism
- Rating and review system
- Dispute resolution module
- Administrative moderation and analytics

7.2 Non-Functional Requirements

- Scalability or large user base
- High security and data privacy
- High availability and reliability
- User-friendly interfaces
- Performance efficiency

8 Proposed Modular Decomposition

Team	Microservice / Module	Description
Team 1	User Authentication & Authorization Service	Login, signup, role-based access (freelancer, employer, admin), JWT/session handling
Team 2	Freelancer Profile Management Service	Freelancer profiles, skills, experience, portfolio metadata
Team 3	Employer Profile & Company Service	Employer registration, company profiles, verification status
Team 4	Skill Assessment Service	Online tests, quizzes, coding tasks, scoring logic
Team 5	Skill Verification & Certification Service	Certificate issuance, validation, expiry management
Team 6	Job Posting & Discovery Service	Job creation, search, filters, categories
Team 7	Job Matching & Recommendation Engine	Match freelancers to jobs using skills, ratings, availability

Team 8	Payment & Escrow Service	Secure payments, escrow holding, release on job completion
Team 9	Rating & Review Service	Freelancer and employer ratings, feedback storage
Team 10	Dispute Resolution Service	Complaint handling, evidence upload, admin decisions
Team 11	Notification & Messaging Service	Email, in-app notifications, system alerts
Team 12	Admin Dashboard & Moderation Service	User management, content moderation, analytics
Team 13 (Optional)	Reporting & Analytics Service	Platform usage, revenue reports, performance metrics

9 Budget Estimate and Financial Analysis

A preliminary estimate of costs for the entire project is PKR 6,000,000, covering development, testing, deployment, and initial operational setup. This estimate is based on a three-month development period, with the project manager working approximately 20 hours per week and development teams working approximately 60 hours per week.

The staff project manager is valued at PKR 2,000 per hour, while other team members (backend, frontend, QA, DevOps, and UI/UX) are valued between PKR 1,300–1,500 per hour, reflecting conservative industry-equivalent rates for academic estimation.

The total cost also includes PKR 1,550,000 for cloud infrastructure, security mechanisms, payment gateway integration, and essential software tools. After project completion, annual maintenance and support costs are estimated at PKR 1,800,000.

Financial Justification (Assumption-Based)

The platform is expected to generate revenue through:

- Skill verification fees
- Premium freelancer subscriptions
- Employer job posting and subscription fees

- Institutional and training partnerships

Based on conservative assumptions, if approximately 20,000 freelancers use paid verification services annually and a small percentage of users subscribe to premium services, the platform can generate PKR 50,000,000 per year in gross revenue.

Even after accounting for annual maintenance costs, the platform is expected to recover its initial development cost within the first year of operation, making it financially viable. These estimates demonstrate that the project is economically justified and sustainable in the long term.

10 Schedule Estimate

The project is expected to be completed within 3 months, with the following schedule:

- Requirements & Planning: 2 weeks
- System Design & Architecture: 2 weeks
- Parallel Module Development: 6 weeks
- Integration & Testing: 2 weeks
- Deployment & Demonstration: 1 week

This schedule allows incremental development by multiple teams and supports full integration by the end of the semester.

11 Potential Risks

This project carries several risks:

- Low user adoption, especially in the early stages
- Technical challenges in implementing fair and reliable skill assessments
- Payment gateway integration issues
- Security and data breach risks, including unauthorized access or data leaks
- Integration delays, due to dependency between modules developed by different teams

Mitigation strategies include phased rollout, modular design, extensive testing, and continuous user feedback.

12 Exhibits

Exhibit A. Cost–Benefit Summary for National Freelance Platform

Category	Amount (PKR)
Initial Development Cost	6,000,000
Annual Maintenance Cost	1,800,000
Estimated Annual Revenue	50,000,000
Net Annual Benefit (After Maintenance)	48,200,000
Estimated Payback Period	Within 1 year

Assumptions:

- 20,000 freelancers use paid verification annually
- Limited premium subscriptions and employer services
- Conservative pricing model
- Revenue begins after system deployment